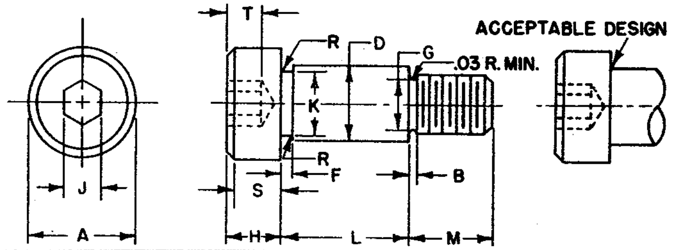
This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



NOMINAL SIZE		1/4		5/16		3/8		1/2		5/8	
THREADS		#10-24NC-3A		1/4-20UNC-3A		5/16-18UNC-3A		3/8-16UNC-3A		1/2-13UNC-3A	
D - SHOULDER DIAMETER	MAX. MIN.	.2480 .2460		.3105 .3085		.3730 .3710		.4980 .4960		.6230 .6210	
A - HEAD DIAMETER	MAX. MIN.	.375 .367		.437 .429		.562 .553		.750 .739		.875 .863	
H - HEAD HEIGHT	MAX. MIN.	.187 .182		.219 .213		.250 .244		.312 .306		.375 .368	
S - HEAD SIDE HEIGHT	MAX. MIN.	.1641 .1589		.1899 .1839		.2155 .2091		.2668 .2594		.3181 .3097	
J - SOCKET WIDTH ACROSS FLATS	MAX. MIN.	.1270 .1250		.1582 .1562		.1895 .1875		.2520 .2500		.3155 .3125	
T - DEPTH OF SOCKET	MAX. MIN.	.1365 .090		.160 .120		.183 .151		.2295 .182		.276 .245	
M - LENGTH OF THREADS		3/8		7/16		1/2		5/8		3/4	
B - WIDTH OF NECK		1/16		5/64		5/64		3/32		1/8	
G - DIAMETER OF NECK	MAX. MIN.	.145 .136		.196 .185		.252 .240		.307 .294		.417 .400	
F - WIDTH OF NECK		3/32		3/32		3/32		3/32		3/32	
K - DIAMETER OF NECK	MAX. MIN.	.243 .240		.305 .302		.368 .365		.493 .490		.618 .615	
R - RADIUS	MAX. MIN.	.020 .015		.020 .015		.025 .020		.025 .020		.030 .025	
* L - SHOULDER LENGTH		DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN
3/8		MS 16637-1									
1/2		-2		MS 16637-8		MS 16637-16				MS 16637-38	
5/8		-3		-9		-17				-39	
3/4		-4		-10		-18		MS 16637-28		-40	
7/8										-41	
1.0		-5		-11		-19		-29		-42	
1-1/4		-6		-12		-20		-30		-43	
1-1/2		MS 16637-7		-13		-21		-31		-44	
1-3/4				-14		-22		-32		-45	
2.0				MS 16637-15		-23		-33		-46	
2-1/4						-24		-34		-47	
2-1/2						-25		-35		-48	
2-3/4						-26		-36		-49	
3.0						MS 16637-27		MS 16637-37		MS 16637-50	

FED. SUP CLASS
5305

APPROVED **18 Dec. 1958** REVISED

P.A. NORD Other Cust ORD	TITLE SCREW, SHOULDER, SOCKET HEAD, HEXAGON ALLOY STEEL, UNCOATED, UNC-3A
PROCUREMENT SPECIFICATION MIL-S-21472 STYLE 1 CL. 1	SUPERSEDES:

MILITARY STANDARD

MS-16637

SHEET 1 OF 2

DD FORM 1 SEP 57 672-1

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-23T13:59Z
Check the source to verify that this is the current version before use.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

NOMINAL SIZE		3/4	1.0
THREADS		5/8-11-UNC-3A	3/4-10-UNC-3A
D - SHOULDER DIAMETER	MAX. MIN.	.7480 .7460	.9980 .9960
A - HEAD DIAMETER	MAX. MIN.	1.000 .987	1.3125 1.2970
H - HEAD HEIGHT	MAX. MIN.	.500 .492	.625 .616
S - HEAD SIDE HEIGHT	MAX. MIN.	.4341 .4251	.5366 .5258
J - SOCKET WIDTH ACROSS FLATS	MAX. MIN.	.3780 .3750	.5030 .5000
T - DEPTH OF SOCKET	MAX. MIN.	.369 .307	.462 .370
M - LENGTH OF THREADS		7/8	1.0
B - WIDTH OF NECK		9/64	5/32
G - DIAMETER OF NECK	MAX. MIN.	.526 .507	.642 .620
F - WIDTH OF NECK		3/32	1/8
K - DIAMETER OF NECK	MAX. MIN.	.743 .740	.993 .990
R - RADIUS	MAX. MIN.	.030 .025	.035 .030

* L - SHOULDER LENGTH	3/4 DASH NO.	3/4 FIIN	1.0 DASH NO.	1.0 FIIN
1.0	MS 16637-51			
1-1/4	-52			
1-1/2	-53		MS 16637-60	
1-3/4	-54		-61	
2.0	-55		-62	
2-1/4	-56		-63	
2-1/2	-57		-64	
2-3/4	-58		-65	
3.0	MS 16637-59		-66	
3-1/4				
3-1/2			-68	
3-3/4				
4.0			MS 16637-70	

MATERIAL: STEEL ALLOY, MINIMUM YIELD STRENGTH 0.2 PERCENT OFFSET 140,000 PSI IN ACCORDANCE WITH THE PROCUREMENT SPECIFICATION.

HARDNESS: ROCKWELL C36-43.

PROTECTIVE COATING: UNCOATED.

THREADS: THE THREADS SHALL BE IN ACCORDANCE WITH SCREW-THREAD STANDARDS FOR FEDERAL SERVICES, HANDBOOK H-28

*SHOULDER LENGTH TOLERANCE: THE ALLOWABLE TOLERANCE ON THE LENGTH (L) UNDER THE HEAD SHALL BE ±.005.

HEAD: PLAIN OR KNURLED HEADS ARE ACCEPTABLE.

DIMENSIONS: IN INCHES UNLESS OTHERWISE SPECIFIED.

NOTES:
(A) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
(B) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS.

FED. SUP CLASS
5305

APPROVED **18 Dec. 1958** REVISED

P.A. NORD
Other Cust
ORD

TITLE
SCREW, SHOULDER, SOCKET HEAD, HEXAGON
ALLOY STEEL, UNCOATED, UNC-3A

MILITARY STANDARD

MS-16637

PROCUREMENT SPECIFICATION
MIL-S-21472 STYLE 1 CL. 1

SUPERSEDES:

SHEET 2 OF 2

DD FORM 1 SEP 57 672-1 PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T13:59Z
Check the source to verify that this is the current version before use.